ERICK ARMANDO RODRÍGUEZ BÁRCENA

SOFTWARE ENGINEER | QA ANALYST

55 40 77 90 49 | México, Querétaro | Portfolio | linkedin.com/in/erickbarcena7 | erickbarcena00@gmail.com

PROFESSIONAL SUMMARY

Mechatronic Engineer with 9+ years of experience delivering high technical engineering solutions by contributing across 22+ impactful projects for diverse international clients. With expertise in full stack development, automation engineering and quality assurance, I'm adept at creating innovative technology solutions, developing AI-driven automation strategies, leading quality assurance procedures, managing cross-functional teams, and collaborating with stakeholders and IT colleagues for business success.

WORK EXPERIENCE

Senior Product Support Engineer | Yalo.ai June 2025 – Aug 2026

Playing a key role in strengthening Yalo's AI conversational platform by designing, maintaining, and optimizing conversational and voice AI agents for over 20 international clients, enabling scalable customer engagement across Meta communication platforms.

- Developed AI-driven solutions that optimized internal workflows, automated client operations, enhanced internal tools, supported AI agents, integrated new applications into existing systems, and delivered root cause analysis for production incidents.
- Built internal scripts and automation tools that increased incident resolution efficiency by automating manual workloads.
- Delivered L2/L3 engineering support across distributed systems, resolving internal and external escalated incidents by analyzing AWS Lambdas and Google Cloud Functions, applying targeted code fixes, and restoring service reliability.
- Refactored and enhanced production services written in Lua, TypeScript, Python and Go to resolve escalated support cases, managing incident lifecycles through Zendesk and ensuring stable system performance.
- Designed and implemented AI self-service tools and automation scripts using MCP and LLM technologies, enabling clients to handle common requests independently while reducing support ticket volume and response times across client operations.
- Optimized and maintained webhook-based integrations that automated event-driven communication between messaging platforms and backend microservices, enabling real-time synchronization and improving operational responsiveness.
- Supported the integration and stability of AI-powered conversational agents interacting with enterprise systems such as Salesforce, SAP, and custom ERPs, ensuring seamless communication through WhatsApp Business API and Messenger.
- Performed end-to-end troubleshooting across Meta communication platforms, ensuring service stability and peak performance.
- Technical Skills: Lua, TypeScript, Python, Node.js, REST APIs, AWS, Go, MongoDB, GCP, Docker, Datadog, Zendesk.

Lead Quality Automation Engineer | Apesa Software. Feb 2025 – June 2025.

Within a single quarter, I automated the company's key financial products, including SPEI and CETES, by designing and implementing an automation framework integrated with CI/CD pipelines. Hired as the company's Lead QA Engineer as a first automation step for their financial systems. I oversaw the end-to-end automation lifecycle while mentoring a team of 12 engineers.

- Conducted hands-on training sessions and workshops on automation, engineering standards, and testing strategies, strengthening the technical proficiency, productivity and overall autonomy of a 12-member engineering team.
- Designed and implemented a complete testing plan for Angular, .NET Core/.NET Framework, Blazor, and Entity Framework applications that covered the automation of critical systems, ensuring seamless QA integration, within agile delivery processes.
- Established and maintained a scalable automation framework within GitLab CI/CD, implementing custom stages, automated reporting workflows, and notification channels ensuring consistent, automated validation across release cycles.
- Executed API and performance testing through Swagger, Postman, Newman and JMeter, integrating SQL-based data validation and shell scripting across Linux and Windows environments, testing monolithic, hexagonal, and microservice architectures for financial systems processing thousands of daily transactions with significant operational responsibility.
- Defined and enforced automation objectives and KPIs, ensuring measurable improvements to overall quality and test coverage.
- Led the QA automation transformation, shifting the team's manual-testing mindset, establishing best practices, documentation, and long-term automation standards that continued beyond my tenure.
- Technical Skills: SQL, Playwright, Angular, .Net Core/ .Net Framework, SonarQube, Swagger, Blazor, Entity Framework, JMeter, Monday.

Lead Quality Assurance Engineer / Pega System Architect | Rules Cube Jan 2021 – Feb 2025

Certified System Architect with experience in building implementing and quality assuring Pega applications across diverse industries, including supply chain, financial and insurance. Architected scalable enterprise solutions, led testing teams, designed and executed test plans, and leveraged automation, decisioning and ETL (Extract, Transform, Load) strategies to drive impactful project results.

- Delivered 5 high impact projects as a Pega Consultant using Pega's Center out business architecture to produce value driven solutions that drove client success and led to significant project renewals and revenue growth.
- Managed cross functional teams of Pega consultants and QA testers implementing end to end testing strategies to manage the execution of a variety of test cases significantly reducing defect rates and ensuring high quality deliverables.
- Developed a large scale Pega based application for a leading supply chain client capable of handling over 1,000 case requests daily, significantly optimizing business operational efficiency and case resolution management.
- Migrated critical data from client source systems to Pega data models ensuring data integrity and reliability, resulting in smooth post implementation performance with no data related issues and successful operational stability using ETL processes.
- Ensured quality on distinct products for a multinational insurance company, effectively implementing automation/manual testing strategies, Scrum methodology and DevOps practices to ensure product compliance.
- Applied design thinking as a Business Analyst, by participating in project planning and resource/scope definitions.
- Designed testing plans, coordinated testing activities and executed a wide range of tests, including integration, functional, unit, acceptance and regression testing across development, staging and production environments.
- Earned certifications as Pega System Architect, Business Architect and ISTQB Tester, demonstrating a commitment to professional growth and technical excellence while applying these skills to drive impactful results in complex projects.
- Technical Skills: JavaScript, SQL, PRPC, Java, Playwright, ETL, API Integrations, DevOps, CI/CD, Scrum, TDD, BDD, Jenkins.

Front-End Software Engineer | Global Human Capital April 2020 – Dec 2020

Responsible of the front end releases for a large scale suite of applications, successfully delivering over 20 responsive landing pages, translating high fidelity designs into responsive pixel perfect interfaces, fetching and rendering dynamic data for real time content updates and ensuring compatibility across multiple devices and screen sizes.

- Developed a key layer of the suite's front-end codebase, producing dynamic responsive landing pages that fulfilled business requirements and directly impacted customer acquisition and product revenue success.
- Rendered dynamic content using JSP to fetch API-driven data and JavaScript to add animations and interactions.
- Improved code performance by implementing optimization techniques such as code splitting, lazy loading, reusable and scalable web components, structured code architecture, asset compression, and SEO best practices to enhance page speed and efficiency.
- Proactively addressed front end defects, leading to a substantial decrease in user interface ticket volume.
- Significantly contributed to the development of the suite's front-end releases, ensuring seamless integration with PHP Laravel server-side components and optimizing data processing by collaborating with back-end developers and DBAs.
- Implemented Apache web server configurations for efficient application deployment and environment management.
- Improved team efficiency by creating technical documentation and streamlined workflows, improving knowledge transfer time, code standards, incident handling, client-server component integrations, and asset management.
- Participated in regular code reviews and performance assessments, while following Kanban methodology for task management, GitHub for version control and CI/CD pipelines for automated builds, deployments and testing.
- Technical Skills: JavaScript, TypeScript, Jest, JSP, GIT, RESTful API Integrations, CSS, Webpack, DevOps, CI/CD, Figma.

Full-Stack Software Engineer | NEORIS May 2019 – April 2020

- Developed and maintained a banking web application for a large-scale financial institution, delivering dynamic releases and building reusable web components using JavaScript, TypeScript, SQL, .NET and React.
- Diagnosed and resolved complex server-side defects including email delivery failures, data processing errors, logic inconsistencies and system edge cases, employing C# while implementing error handling with JavaScript.
- Executed optimized queries in SQL Server for efficient data retrieval, continuous monitoring and application debugging, strengthening system reliability, operational performance, and minimizing data inconsistencies.
- Utilized Microsoft SharePoint, leveraging internal farms for efficient data management, integrated with Azure high performance computing resources to handle demanding workloads, improving overall system scalability.
- Deployed and managed virtual machines (VMs) using virtualization platforms such as Hyper V and VirtualBox, ensuring stable and secure environments for development and staging, while leveraging GitHub for version control.
- Managed network computing tasks including IP configuration, IIS setup, and network protocol administration.
- Conducted technical meetings and demos, offering precise communication with stakeholders and team members throughout the project lifecycle, facilitating informed decision-making, ensuring alignment and driving project success.
- Technical Skills: JavaScript, React, TypeScript, C#, SQL, GIT, RESTful APIs, SharePoint, Postman, Azure, DevOps (CI/CD).

MIS Solutions Engineer / Premier Technical Analyst | HSBC Financial Group. June 2018 – May 2019

- Developed Management Information System (MIS) solutions to streamline data processing and automate reporting workflows for the premier department, utilizing command line interfaces (CLIs), SQL, and Excel VBA for data management.
- Automated repetitive reporting tasks using VBA scripts, generating custom data driven reports with Power BI.
- Delivered prompt and accurate 1st and 2nd level technical support to premier level banking clients, providing personalized solutions and proactive problem solving to high priority customers, ensuring elevated levels of satisfaction.
- Maintained a high level of compliance with bank procedures, metrics and customer needs, providing tailored solutions that consistently received outstanding feedback, strengthening client trust and retention of high priority clients.
- Utilized remote desktop tools to identify, diagnose and resolve technical issues on support team devices, implementing root cause analysis and preventive measures to ensure seamless operations and minimize disruptions.
- Diagnosed and implemented technical solutions to resolve complexities with banking applications, operational processes, customer support tickets and device assistance, ensuring accurate solutions, minimal disruption and service reliability.

Bilingual Customer Solutions Executive | Teleperformance Oct 2017 – June 2018

- Provided in-depth technical troubleshooting for complex incidents across banking, telecommunications, and logistics sectors.
- Delivered accurate, results-oriented and client-oriented solutions, resulting in enhanced levels of customer satisfaction.
- Exceeded QA expectations by consistently meeting KPIs, earning the top performer award and senior acknowledgment.

EDUCATION

Master's in Business Management & Quality Engineering for Manufacturing Systems | Universidad Tecnológica de México | . Present

- Complete specialization in business management for manufacturing and technology environments, emphasizing automation, process optimization, quality systems, and continuous improvement applied to software, industrial and engineering contexts.

Postgraduate Diploma in Manufacturing, Technology and Quality Systems | Universidad Tecnológica de México. | 2022-2023

- Focused on advanced methodologies for optimizing manufacturing systems, integrating software development and automation with quality assurance, strategic leadership, and continuous improvement in business operations.

Bachelor's in Mechatronic Engineering | Universidad Tecnológica de México | 2019-2022

- Specialized in automation systems, software development, electronics, industrial control, manufacturing processes, and robotics.

Bachelor's in Mechatronic Engineering and Production |. Universidad Iberoamericana | 2017 - 2019

- Focused on the integration of mechanical, electronic, and computing systems for automation, control, and system integration.
- Participated in multidisciplinary engineering projects that promoted new initiatives for the university's engineering association.

TECHNICAL SKILLS

- **Programming & Scripting:** JavaScript (ES6+), TypeScript, C, Lua, C#, Python, Bash, Java, VBA.
- **Frameworks & Libraries:** React, Next.js, Node.js, Express.js, Angular, Blazor, .NET, Playwright, Jest, GSAP, Cypress, React Native, Tailwind, Gatsby, Bootstrap, Framer Motion, Django, ASP.NET Core, GTK, Material UI, Three.js.
- **Databases:** MySQL, PostgreSQL, SQL Server, MongoDB.
- **Cloud & DevOps:** AWS, GCP, Docker, GitHub Actions, GitLab CI/CD, Jenkins, Kubernetes (K8s), Firebase, Snyk.
- **Testing & QA Tools:** Postman, Swagger, Newman, SonarQube, Playwright, Cypress, API Testing, JMeter, Jira, Monday, Confluence.
- **Automation & Scripting Tools:** Power BI, Excel VBA, Bash, n8n, Lua Scripting, MCP & LLM Integration, Webhooks, Cursor.
- **Data & Analytics:** Power BI, Tableau, SQL Querying, AWS (EC2, Lambda, S3, CloudWatch), BigQuery, GCP, NumPy, Datadog.
- **Enterprise Systems & Tools:** Pega PRPC, Zendesk, ServiceNow, SAP, ERPs, SharePoint, Salesforce, HyperV.
- **Engineering Skills/Tools:** PLC Programming, Microcontrollers (Arduino, PIC, STM32, Raspberry Pi), MATLAB & Simulink, PID Systems, Autodesk Inventor, AutoCAD, SolidWorks, Multisim, Proteus, Claude Code, Electronics Design & Simulation, Sensors & Actuators, Instrumentation (Oscilloscope, Signal Generator, DAQ Systems), Lean Manufacturing, Six Sigma, Process Automation, Control Theory, Industrial Control Systems.
- **Web & UI/UX:** Figma, Adobe Illustrator, JSP, Webpack, HTML5, CSS3, WordPress, ESLint, GraphQL, RESTful APIs.
- **Soft Skills:** Leadership, Mentorship, Root Cause Analysis, Agile (Scrum, Kanban), Continuous Improvement, Cross-functional Collaboration.

Languages

- English – Advanced (C1)
- French – Basic (A1)
- Spanish – Native (C2)

Certifications

- ISTQB Foundation Certified | ISTQB International Board
- Certified Pega System Architect | Pearson VUE
- Certified Pega Business Architect | Pearson VUE
- CS Intro to Computer Science | Ongoing, Harvard
- Certified JavaScript Developer | Verified by Talently
- Certified React Developer | Verified by Talently

Awards & Accomplishments

- Successfully delivered high-quality impactful products across diverse technical roles showcasing skills and results.
- Proficiently mentored and led groups of engineers, establishing clear objectives and achieving measurable outcomes.
- Implemented AI-driven automation solutions that increased efficiency and reduced manual workload across key processes.
- Engaged in initiatives as a member of the Mechatronics, Telecommunications, and Biomedical Engineering Association.